



TSMC 0.18um USB 2.0 I/O Library

tpd973g_usb2

Silicon Report

v250b

2005/12/9

Taiwan Semiconductor Manufacturing Co., Ltd.

TSMC 0.18um USB 2.0 I/O Library

tpd973g_usb2

Silicon Report

Copyright© 2005, Taiwan Semiconductor Manufacturing Company Ltd.
All Rights Reserved

No part of this publication may be reproduced in whole or part by any means without the prior written consent.

NOTICE

Taiwan Semiconductor Manufacturing Company Ltd. Reserves the right to make changes in the contents of this document without notice. No responsibility is assumed by Taiwan Semiconductor Manufacturing Company Ltd. For any infringements of patents or other rights of the third parties that may result from its use. Taiwan Semiconductor Manufacturing Company Ltd. Assumes no responsibility for any error that appears in this document.



TABLE OF CONTENTS

<u>Table of Contents</u>	<u>i</u>
<u>Revision History</u>	<u>1</u>
<u>1. Function Diagram</u>	<u>2</u>
1.1 Functional Blocks of USB 2.0 Transceiver	2
1.2 Functional Blocks of USB 2.0 Test Chip	3
<u>2. Evaluation Board</u>	<u>4</u>
<u>3. High Speed Measurement</u>	<u>5</u>
3.1 High Speed On-Chip Termination	5
3.2 High Speed Signaling Rise and Fall Times	6
3.3 High Speed Signaling Eye Patterns	7
3.4 High Speed Squelch State Measurement	9
3.5 High Speed Current Consumption	10
<u>4. Full Speed Measurement</u>	<u>11</u>
4.1 Full Speed Signaling Rise and Fall Times	11
4.2 Full Speed Signaling Eye Patterns	12
<u>5. ESD Performance</u>	<u>13</u>
<u>6. Conclusion</u>	<u>14</u>



REVISION HISTORY

Version	Date	Special Note
100	2003/10/7	1. First release
110	2004/2/17	1. Modify the description of high speed current consumption 2. Remove the description of high speed transmission lose packet due to have solved this issue
250b	2005/12/9	Provided dedicated silicon report for tpd973g_usb2_250b library

1. FUNCTION DIAGRAM

1.1 Functional Blocks of USB 2.0 Transceiver

Fig. 1 shows the functional blocks of the USB 2.0 transceiver macro cell. According to the “Universal Serial Bus Specification Version 2.0”, the following chapters describe the silicon measurement of the electrical parameters with respect to the USB 2.0 transmitter and receiver. The eye-patterns are also included for both high-speed and full-speed modes.

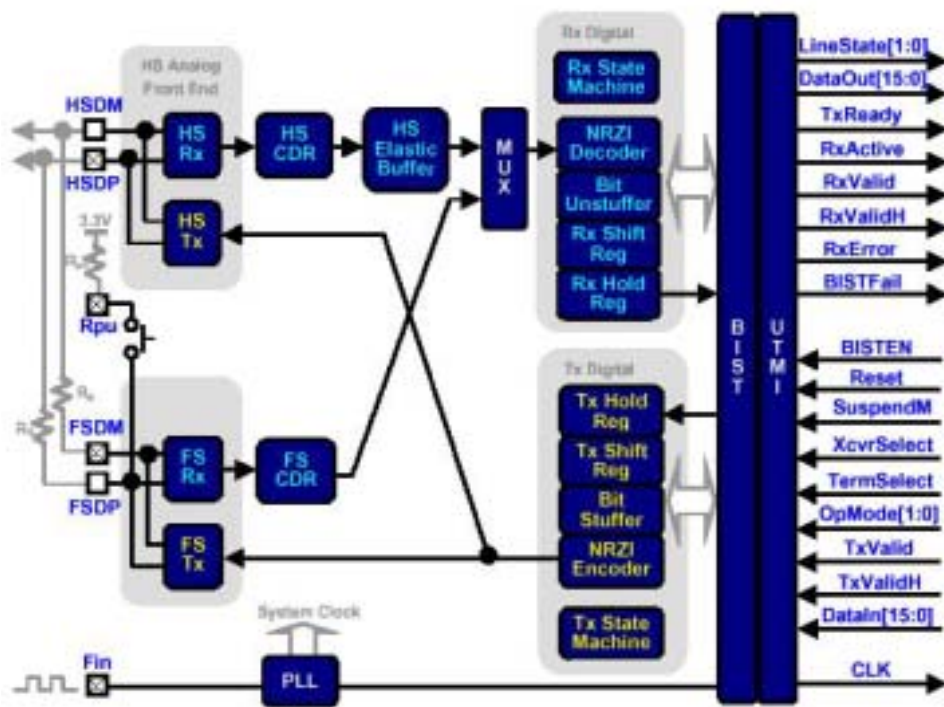


Fig. 1 Function blocks of USB 2.0 transceiver (16-bit mode)

1.2 Functional Blocks of USB 2.0 Test Chip

Fig. 2 shows the functional diagram of the USB 2.0 test chip for *tpd973g_usb2*. The test chip is designed with “TSMC 0.18um CMOS Logic General Purpose 1P6M” process. The PLL is used to generate the internal high frequency multi-phase clock, 480MHz for examples. The FPGA is used to generate 56 bytes Test_Packet which is described in section 7.1.20 of “Universal Serial Bus Specification Revision 2.0” and this enables the testing of rise and fall times, eye patterns, jitter, and any other dynamic waveform specifications. Besides, the CDR can recover the clock signal from the received data stream. The internal high frequency signals can be sent out by the USB-like transmitter to get the real-time high-speed data.

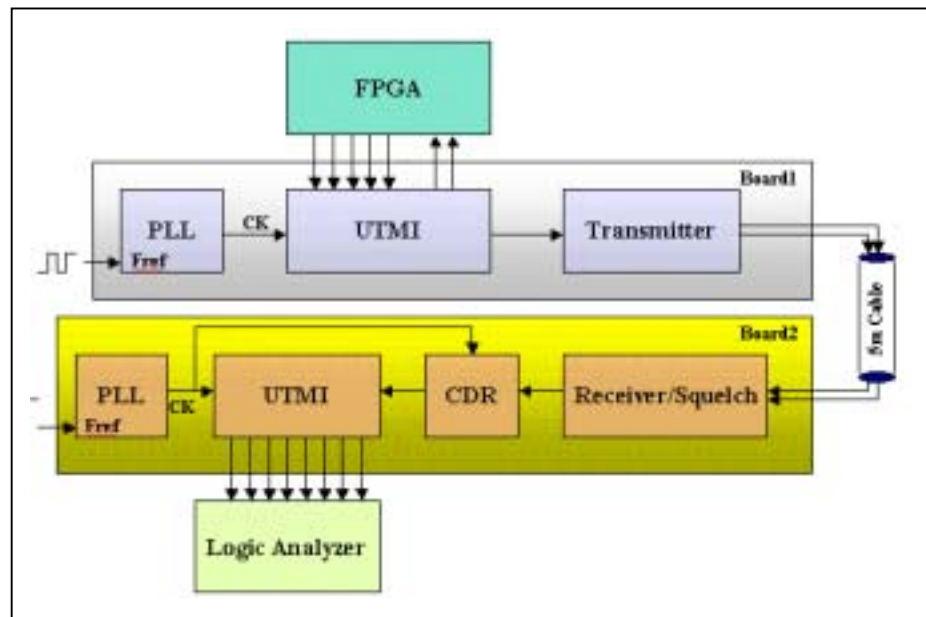


Fig. 2 Function blocks of USB 2.0 test chip

2. EVALUATION BOARD

Fig. 3 shows the USB 2.0 evaluation board with the test chip for *tpd973g_usb2*. The test chip is packaged with 208-pin PQFP package and the evaluation board is designed with 4-layer FR4 PCB. The on-board regulator is included to provide stable power and reduce ground noise. Please note that our USB 2.0 transceiver is used for device application only, so you can see that USB B-receptacle is located at the lower boundary of the board and the A-receptacle is not shown in the photograph.



Fig. 3 Photo of USB 2.0 evaluation board with test chip

3. HIGH SPEED MEASUREMENT

3.1 High Speed On-Chip Termination

The USB 2.0 specified that the DC resistance from D+ or D- to the device ground is required to be $45\Omega \pm 10\%$ when measured without a load. Fig. 4 and Fig. 5 show the TDR measurement result and photo.

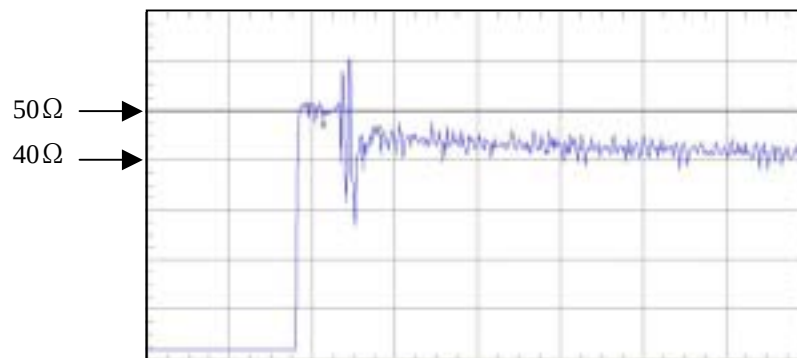


Fig. 4 TDR measurement result for 45Ω termination

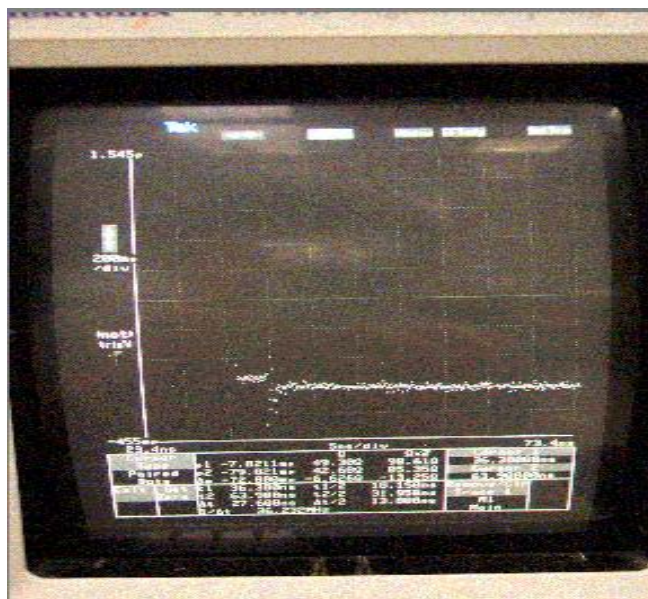


Fig. 5 Photo of TDR measurement

3.2 High Speed Signaling Rise and Fall Times

The transition time of a high-speed driver must not be less than the specified minimum allowable differential rise and fall time (ie 500ps). Transition times are measured when driving a reference load of 45Ω to ground on D+ and D-. Fig. 6 shows a recommended “Transmitter Test Fixture” for performing these measurements and Fig. 7 shows the measurement result. You can see that the rise and fall times are about 550ps.

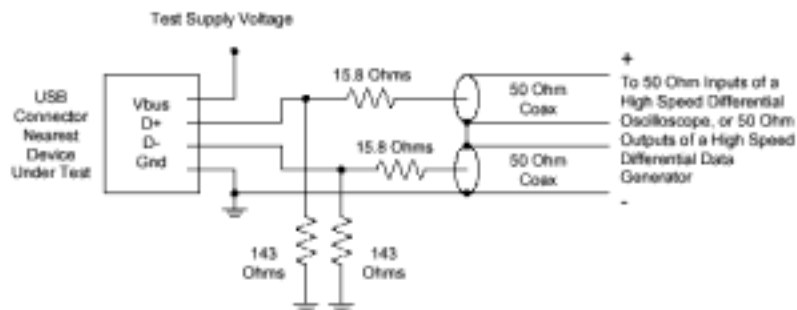


Fig. 6 Transmitter/Receiver test fixture

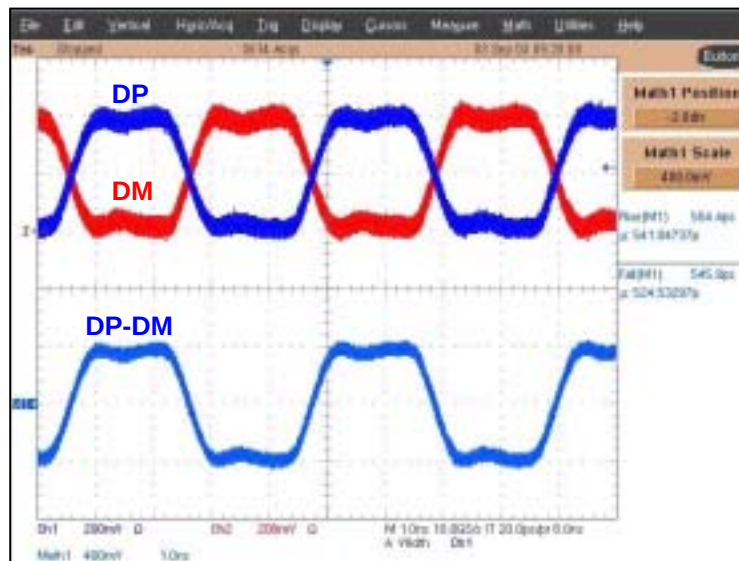


Fig. 7 Waveforms of high speed signaling with rise/fall time measurement

3.3 High Speed Signaling Eye Patterns

In order to measure the eye patterns, we applied an off-chip pseudo random number generator to generate the transmission pattern randomly. It's easy to observe that the jitter and overshoot /undershoot performance.

For high speed mode signaling, all bits (including the first and last bit of a packet) must meet the eye pattern requirements for timing and amplitude. Fig. 8 shows the transmit waveform requirements and Fig. 9 shows the receiver sensitivity requirements. The measurement results are shown in Fig. 10 and Fig.11. The required eye opening is drawn in blue hexagon, and the waveforms should not touch those regions. It's shown that the waveforms meet the requirement. Besides, the peak-to-peak jitter is only 182ps for transmit side and 373ps for receiver side with 5m USB cable connection.

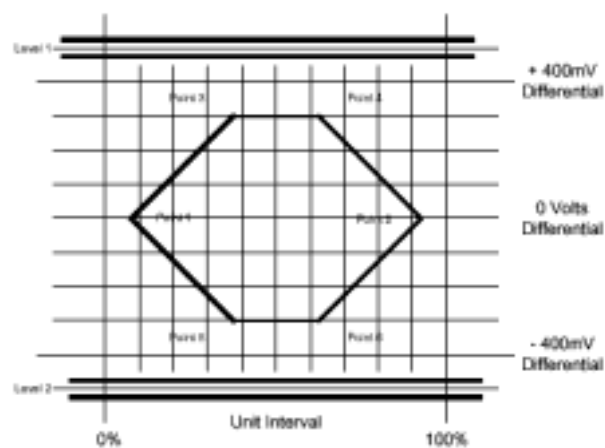


Fig. 8 High speed transmit waveform requirements

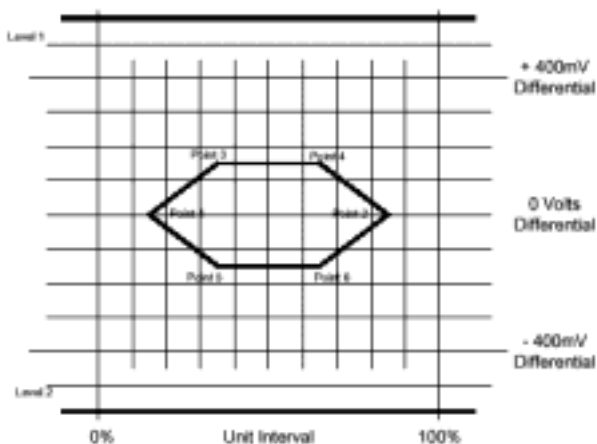


Fig. 9 High speed receiver sensitivity requirements

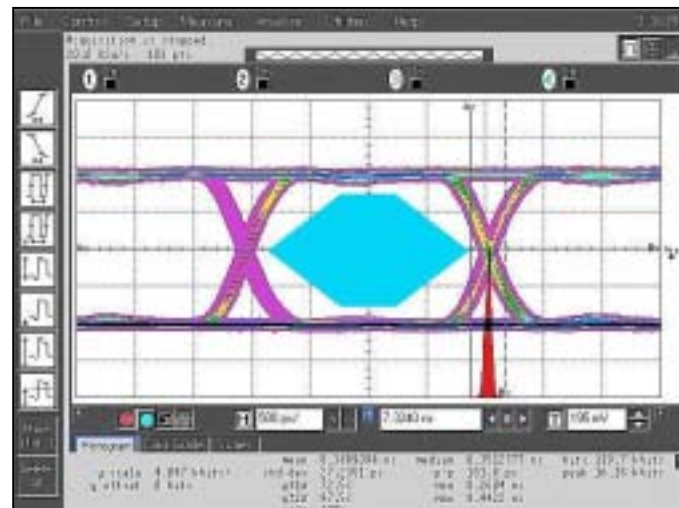


Fig. 10 Eye pattern for transmit side (Near End)

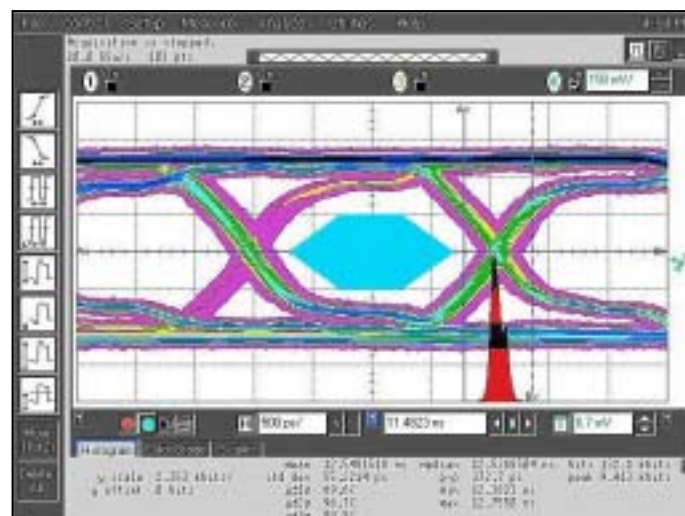


Fig. 11 Eye pattern for receiver side with 5m USB cable connection (Far End)



3.5 High Speed Current Consumption

The following table lists the measurement result of current consumption for our USB 2.0 transceiver macro. The current consumption of the whole macro is only 68mA and the total power consumption is only 204.9mW. The major part comes from “PLL” “High Speed Transmitter” and “High Speed Receiver/Squelch” blocks. Besides, in power down mode, the suspend leakage current is less than 5uA. It is very small and can meet specification.

	CURRENT (MA)	POWER (MW)
Core Power (1.8v)	13	23.4
I/O Power (3.3v)	55	181.5
Whole Macro	68	204.9
Suspend Leakage	< 5uA	

4. FULL SPEED MEASUREMENT

4.1 Full Speed Signaling Rise and Fall Times

According to the USB 2.0 specification, Fig. 14 shows the full-speed test configuration for rise time and fall time measurement. The load capacitance (C_L) is set to be 50pF and the series resistance (R_s) value is for reference only. The measured waveforms of D+ and D- are shown in Fig. 15. The rise and fall time are about 9ns and can meet specification.

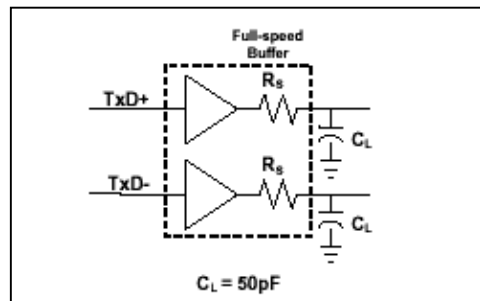


Fig. 14 Test configuration for AC timing of full-speed mode transmission

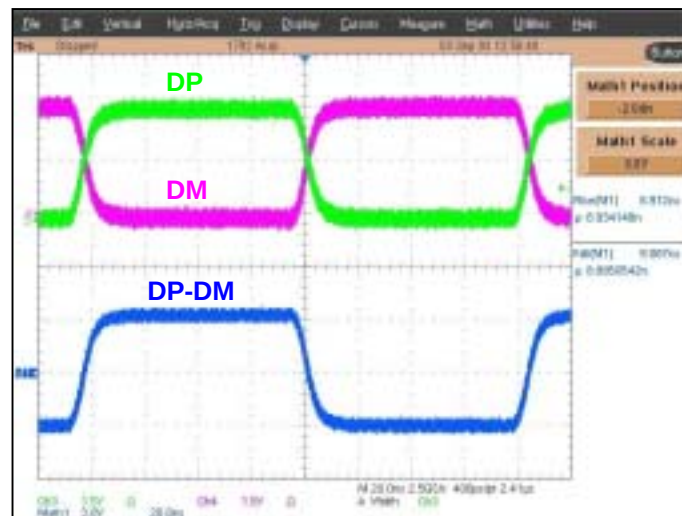


Fig. 15 Waveforms of full speed signaling with rise/fall time measurement

4.2 Full Speed Signaling Eye Patterns

To measure the eye patterns of the full-speed transmission, we also applied an off-chip pseudo random number generator to generate the transmission pattern randomly. Fig. 16 and Fig. 17 show the eye diagrams of full-speed transmission. It's shown that the waveforms can meet the requirements. Besides, the peak-to-peak jitter is small and which is suitable for most applications.

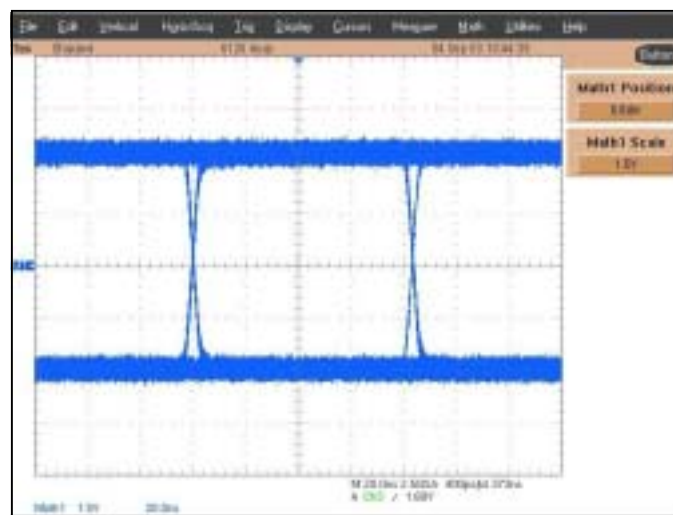


Fig. 16 Eye pattern for transmit side

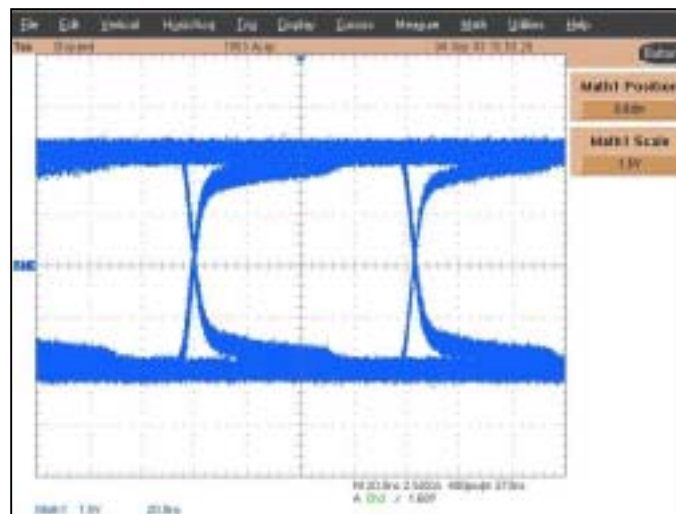


Fig. 17 Eye pattern for receiver side with 5m USB cable connection



5. ESD PERFORMANCE

For ESD (electro-static discharge) test, we performed HBM (human body model) and MM (machine model) that are specified to be HBM 2KV and MM 200V. The *tpd973g_usb2* test chip has been tested under each test mode and passed the ESD specifications shown above. User should follow the USB 2.0 transceiver macro cell placement with power cut cell to optimize the whole chip ESD performance.



6. CONCLUSION

From the key USB 2.0 electrical parameters measured above, we can prove that the library *tpd973g_usb2* functions correctly with silicon. It also passes the ESD test and shows good reliability.

For any further technical questions or support, please contact your TSMC representative or local FAE.